Calculus I Homework 4

Department of Mathematics

Instructions

Show all work and simplify final answers. You may use a calculator for arithmetic only.

Problems

1. Compute the limit, or state that it does not exist.

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$$

P1.

2. Find the derivative of the function.

$$f(x) = x^3 \ln(x) - \frac{4}{x^2}$$

P2.

3. Compute the definite integral and sketch $y = 3x^2 - 4x + 1$ with the area for the integral shaded.

$$\int_0^2 (3x^2 - 4x + 1)\, dx$$

Figure

P3.

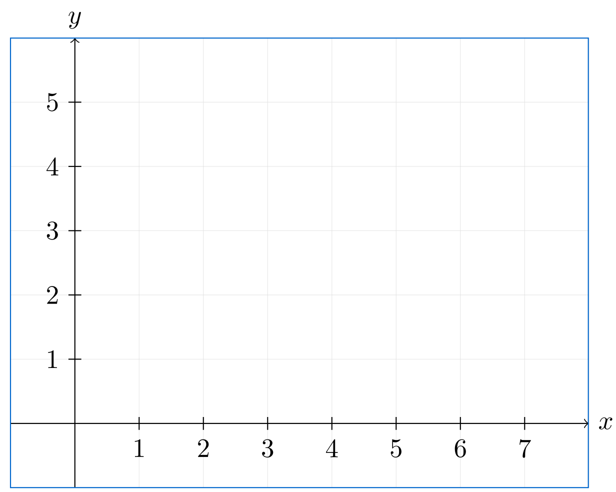
4. Let $f(t)$ be defined by

$$f(t) = \begin{cases} 2 & 0 \le t < 2, \\ 2 - t & 2 \le t < 4, \\ t - 4 & 4 \le t \le 6. \end{cases}$$

Define

$$F(x) = \int_0^x f(t)\,dt.$$

On the axes below, sketch the graph of $F(x)$ for $0 \le x \le 6$.



5. Use the Fundamental Theorem of Calculus to find the derivative.

$$G(x) = \int^{x} \sqrt{1+t^4}\,dt$$